Mohsen Dirbaz

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PROFILE

A well-rounded Chemical Engineer with 5+ years in-depth knowledge of process design, modeling, and simulation in oil & gas industries, ASPEN Plus, HYSYS, PFD's, P&ID's, and heat and material balances. Strong foundation in process engineering with hands-on experience in core unit operations (e.g., distillation, heat exchange, reactors) and supporting equipment such as pumps, compressors, and separation systems. A solid combination of academic excellence, technical proficiency, and practical project experience, complemented by a proactive, self-motivated, and detail-oriented approach to individual and collaborative endeavors. Served as a Process Engineer at L&T Technology Services, where responsibilities included review of PFDs and PIDs, fluid flow calculations, and coordination with equipment vendors and fabrication partners. Technical capabilities span advanced simulation and programming tools such as Aspen Plus, HYSYS, MATLAB, Mathematica, Python, JavaScript, NumPy, and Jupyter notebooks. Recognized with multiple honors and awards, including Teaching Assistant of the Year at Illinois Institute of Technology and competitive graduate assistantships. Active participation in professional organizations such as AIChE and SPE, as well as leadership roles in academic associations.

EDUCATION

Chemical Engineering | Ph.D.
Illinois Institute of Technology (IIT)
2012 - 2020 CHICAGO, US
GPA: 3.6 (2012-2020)
Dissertation: "A Neural Network Based Model for Biomass Gasification in Fluidized bed"Researched gasification and emission control technologies relevant to IGCC systems, including biomass and coal gasification in fixed, entrained, and fluidized bed reactors.Investigated intermediate purification and treatment steps—such as sulfur compound removal, trace metal abatement, COS hydrolysis, and carbon capture and other enabling technologies that enhance technical and economic viability of power generation and chemicals production.Conducted research on the environmental impacts of hydraulic fracturing and evaluated emerging frac water remediation strategies—including physico-chemical and biological treatment pathways, process models (e.g. water chemistry models for calculation of PH and precipitation reactions, mass transfer models for oxygen modeling and other gas/liquid interactions) with a focus on sustainable water management practices.

Natural Gas Engineering | M.S.
Texas A&M University-Kingsville (TAMUK)
2010 - 2012 KINGSVILLE, US
GPA: 3.9 (2010-2012)
Dissertation: "Evaluation of Formation Damage in Horizontally Fractured Wells in Low Permeability Reservoirs"Performed advanced process simulation and optimization of industrial gas processing systems using HYSYS software, including acid gas sweetening, dehydration, and fractionation units.Designed and optimized compressor station configurations for a 1,800-mile natural gas transmission pipeline to ensure efficient flow and minimize pressure losses.Performed preliminary economic analysis and material selection to determine optimal pipe diameter, grade, and specification based on maximum pressure and performance requirements.Conducted specialized research in offshore petroleum engineering, focusing on subsea well completion technologies, cementitious material rheology, and enhanced oil recovery screening methodologies.

EDUCATION

Petroleum Engineering | B.Sc.
Sharif University of Technology (SUT)
2005 - 2010 TEHRAN, IRAN
(2005-2010)
Gained hands-on experience in PVT lab, drilling lab, reservoir rock, and fluid mechanics lab.Completed a comparative analysis for constant composition expansion and constant volume depletion.

EXPERIENCE

Postdoctoral Fellow
University of Missouri-Columbia
2023 - 2025 COLUMBIA, US

- Developed TEA Space, a platform that facilitates comprehensive techno-economic assessments by integrating dynamic modeling and simulation capabilities in a highly customizable interface.
- Conducted in-depth analysis on feasibility of manufacturing Nitrogen trifluoride, a critical material used by semiconductor foundries for wafer etching and chamber cleaning.
- Presented at AICHE: “Optimally Dispersed Network of Biomass Gasification to Produce Green H2”
- Authored several technical reports:
- NH3 Report:
- Gray Haber-Bosch Process. Advanced Blue Processes.
- Carbon Capture Technologies. Hydrogen Transportation and Storage Options.
- Green Ammonia via Electrochemical Synthesis. Lithium Mediated Ammonia Production.
- H2 Report:
- Hydrogen Production Routes. Electrolysis. Fossil Resources. Biomass.
- Hydrogen Powered Fleet. Expansion Strategies and Market Penetration.
- Biomass Carbon Neutrality. Circular Economy. Life Cycle Assessment.
- Tinkered An Energy-Policy Roadmap, Introducing A Forward-Thinking Renewable Energy Blueprint Built on 6 Pillars of:
- Climate liability (adaptation vs mitigation)
- Electrifying rural economies (tribal, local, state and territorial models)
- Sustainable forestry (management, certification, carbon accounting)
- Biomass utilization (value chain pathway optimization)
- Hydrogen demand (incentive mechanisms)

Energy Initiatives Consultancy
Didas International Group/Iran
2020 - 2023 TEHRAN, IRAN

- Reviewed Regulatory Compliance Status of Flare Treatment Technologies.
- Mapped Silica Processing End Use Applications of Reflective Coatings for Solar Thermal Systems.
- Analyzed Impact of Euro Adoption on the GDP Dynamics of Middle Eastern Countries.
- Investigated Systemic Disparity of Training-Deployment in Engineering Education in US.

Process Engineer
Larsen and Toubro Technology Services Limited
2019 - 2019 ST LOUIS, US

- Acted as key liaison between offshore engineers and clients, guaranteeing project alignment.
- Reviewed PFDs (Process Flow Diagrams) and P&IDs (Process and Instrumentation Diagrams).
- Delivered comprehensive progress reports at various milestones, fostering transparency and client trust.
- Collaborated with suppliers to finalize equipment designs, optimizing project outcomes.

EXPERIENCE

Teaching Assistant
Illinois Institute of Technology
2015–2017 | CHICAGO, US

- Served as Teaching Assistant in Fluid Mechanics, Thermodynamics I/II, and Process design I/II.

Intern
Didas International Group/Iran
2009–2010 | TEHRAN, IRAN

- Prepared technical reports and presented key insights to internal and external stakeholders.

HONORS AND AWARDS

Teaching Assistant of the Year Award at IIT
2016
Recipient of Teaching Assistant of the Year Award at IIT (2016 & 2017).

Graduate Assistantship, Illinois Institute of Technology
2012
Recipient of Graduate Assistantship, Illinois Institute of Technology, Fall 2012.

Graduate Assistantship, University of Texas A&M-Kingsville
2010
Recipient of Graduate Assistantship, University of Texas A&M-Kingsville, Fall 2010.

Full Scholarship, Sharif University of Technology
2005
Recipient of Full Scholarship, Sharif University of Technology, Fall 2005.

AFFILIATION

- Member of American Institute of Chemical Engineers (2014–present).
- Liaison of ChBE Ethics Committee (NSF-funded project, 2018).
- President of Graduate Student Association at ChBE department, IIT (2016–2017).
- Member of Society of Petroleum Engineers (2010–2012).
- Member of executive student committee of the 2nd and 3rd HSE conference at SUT (2009–2010).
- Representative of petroleum engineering students at Sharif University of Technology (2005–2010).

PUBLICATIONS

The complete 42-title collection, including each full descriptive subtitle, begins on the following page and continues through page 7.

COLLECTION

PUBLICATIONS

VIEW ONLINE PUBLICATIONS

42 research titles | Full descriptive subtitles

01 **A Special-Function-Parameterized Heuristic Reduced-Order Model for Thermochemical Conversion Systems**
Dirbaz, M. | In review

02 **Benefit Distribution in Climate Investment: Allocation Architecture and a Techno-Economic Assessment of Distributed Biomass-to-Hydrogen Pathways**
Dirbaz, M. | In review

03 **Constitutional Geometry of Adaptive Systems**
A Formal Theory of Admissibility, State, Evidence, and Governed Transformation

04 **From State Space to Decision Space**
A General Formal Theory of Geometry-Native Computation Under Constraints and Finite Resources

05 **The Calculus of Sufficient Distinction**
Necessary and Sufficient Conditions for Identifiability, Admissibility, and Task-Relative Resolution

06 **Proof-Carrying Optimization**
A Formal Theory of Joint Optimization, Verification, Certification, and Stopping Under Finite Resources

07 **Decomposition Without Loss**
Characterization Theorems for Exact Local Decomposition, Interface Invariants, and Global Decomposition

08 **Jacobian-Metric-Energy Geometry**
A Differential Theory of Sensitivity, Transformation Cost, Invariants, and Constrained Evolution

09 **Thermodynamic Length as Transformational Geometry**
A Formal Theory of Metric Cost, Geodesic Structure, and Optimal Transformation Paths

10 **Temporal Closure as a Coordination Invariant**
A Formal Theory of Validity, Synchronization, Freshness, and Causal Consistency in Time-Indexed Systems

11 **The Geometry of Receptors**
A Formal Theory of Resolution, Distinguishability, Information Capacity, and State-Space Embedding

COLLECTION

PUBLICATIONS

42 research titles | Full descriptive subtitles

12 **Identity Before Inference**
An Impossibility and Characterization Theory of Recognition Under Observational Ambiguity

13 **Admissibility Before Probability**
A Formal Theory of Constrained Hypothesis Spaces, Inference, and Decision

14 **The Choreography of Computational Change**
A Formal Theory of Typed, Timed, Reversible, and Constitutionally Valid State Transitions

15 **Invariant-Driven Computational Dynamics**
A Formal Theory of Event-Native Computation, Collective State, Geometric Invariants, and Closure

16 **Nullification as a Primitive of System Evolution**
A Formal Theory of Absence, Perturbation Propagation, Structural Reorganization, and Invariant Change

17 **A General Theory of Adaptive Repair**
Characterization of Admissible Reconstruction, Identity Preservation, Sufficiency, and Valid Successor States

18 **The Geometry of Evidence**
A Formal Theory of Provenance, Authority, Uncertainty, Distinguishability, and Justified Inference

19 **Finite-Budget Intelligence**
A Formal Theory of Joint Allocation Across Representation, Search, Computation, Verification, Proof, and Stopping

20 **From Observation to Constitutional Action**
A Formal Theory of Evidence-Constrained State Reconstruction and Governed Transformation

21 **The Calculus of Governed Transformation**
A Unified Formalism for Differential Change, Discrete Transitions, Topological Reorganization, and Valid Successor States

22 **Toward a General Theory of Adaptive Systems**
Formal Foundations of Geometry, Identity, Temporal Closure, Evidence, Optimization, and Constitutional Computation

COLLECTION

PUBLICATIONS

42 research titles | Full descriptive subtitles

06 / 07

23 **The Learning Dynamics of Adaptive Systems**
A Formal Theory of Structure Acquisition, Credit Assignment, and Constitutional Plasticity

24 **Causal Inference in Adaptive Systems**
A Formal Theory of Intervention, Counterfactual, and Structural Discovery Under Constitutional Constraints

25 **Information-Theoretic Limits of Adaptive Computation**
A Formal Theory of Entropy Bounds, Rate-Distortion, and Constitutional Capacity

26 **Consensus and Coordination in Distributed Adaptive Systems**
A Formal Theory of Agreement, Byzantine Faults, and Constitutional Synchronization

27 **Scaling Laws and Finite-Size Effects in Adaptive Systems**
A Formal Theory of Asymptotic Behavior, Critical Size, and Constitutional Scaling

28 **Symmetry, Invariance, and Symmetry Breaking in Adaptive Systems**
A Formal Theory of Group Actions, Orbits, and Constitutional Symmetry

29 **Renormalization and Multi-Scale Adaptive Systems**
A Formal Theory of Coarse-Graining, Fixed Points, and Constitutional Universality

30 **Stochastic Adaptive Systems**
A Formal Theory of Noise, Fluctuation, and Constitutional Stability

31 **Fault-Tolerant Adaptive Systems**
A Formal Theory of Error Models, Redundancy, and Constitutional Recoverability

32 **Robust Fault-Tolerant Policies in Reinforcement Learning**
A Formal Theory of Policy Redundancy, Adversarial Fault Tolerance, and the Hand-in-Hand Nature of Adaptivity and Robustness

COLLECTION

PUBLICATIONS

42 research titles | Full descriptive subtitles

33 **Adversarial Robustness and Security in Adaptive Systems**
A Formal Theory of Perturbation, Information Leakage, and Constitutional Integrity

34 **The Thermodynamics of Adaptive Computation**
A Formal Theory of Landauer Bounds, Entropy Cost, and Constitutional Reversibility

35 **Generalization and Sample Complexity in Adaptive Systems**
A Formal Theory of Learning Bounds, Empirical Risk, and Constitutional Regularization

36 **Meta-Adaptation and Learning to Learn**
A Formal Theory of Higher-Order Plasticity, Constitutional Meta-Structure, and Adaptive Adaptation

37 **Open-Ended Evolution and Novelty Generation**
A Formal Theory of Unbounded Adaptation, Structural Innovation, and Constitutional Openness

38 **The Dynamical Flow and Spectral Theory of the Interface Hypersurface**
Installment V: Transport, Conservation, Heat Kernel, Index, and the Semiclassical Bridge

39 **Criticality and Phase Transitions in Adaptive Systems**
A Formal Theory of Order Parameters, Susceptibility, and Constitutional Criticality

40 **Network Topology and Adaptive Capacity**
A Formal Theory of Connectivity, Robustness, and Constitutional Information Flow

41 **The Belief Complex**
A Unified Theory of Adaptive Systems Under Partial Observation

42 **Temporal Logics for Adaptive Systems**
A Formal Theory of Specification, Model Checking, and Constitutional Verification